## Exp No:15 Experiment to understand data preprocessing in Data Science.

## Program:

```
import numpy as np
import pandas as pd

df = pd.read_csv("pre-process_datasample.csv")
print("Original Dataset:") print(df) print()

print("Dataset Info:")
df.info() print()

print("Mode of Country column:")
print(df.Country.mode()) print()



df['Country'] = df.Country.fillna(df.Country.mode()[0])
df['Age'] = df.Age.fillna(df.Age.median()) df['Salary']
= df.Salary.fillna(round(df.Salary.mean()))
print("Dataset after handling missing values:")
print(df) print()

print("One-Hot Encoded Country column:")
print(pd.get_dummies(df.Country))
print()

updated_dataset = pd.concat([pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
axis=1) print("Updated dataset after One-Hot Encoding:") print(updated_dataset)
print() print("Updated Dataset Info:") updated_dataset.info() print()

updated_dataset['Purchased'] = updated_dataset.Purchased.replace(['No', 'Yes'],
[0, 1]) print("Final Preprocessed Dataset:") print(updated_dataset)
```

**Output**

```
Original Dataset:
   Country   Age   Salary Purchased
0    France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2    Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4    Germany  40.0      NaN       Yes
5    France  35.0  58000.0       Yes
6    Spain   NaN  52000.0        No
7    France  48.0  79000.0       Yes
8    Germany  50.0  83000.0        No
9    France  37.0  67000.0       Yes

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9 Data
columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
0   Country    10 non-null     object
1   Age        9 non-null      float64
2   Salary     9 non-null      float64 3   Purchased  10 non-null     object dtypes: float64(2), object(2) memory usage:
    452.0+ bytes

Mode of Country column:
0    France
Name: Country, dtype: object

Dataset after handling missing values:
   Country   Age   Salary Purchased
0    France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2    Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4    Germany  40.0  63778.0       Yes
5    France  35.0  58000.0       Yes
6    Spain  38.0  52000.0        No
7    France  48.0  79000.0       Yes
8    Germany  50.0  83000.0        No
9    France  37.0  67000.0       Yes

One-Hot Encoded Country column:
   France  Germany  Spain
```

```
0    True    False  False
1    False    False   True
2    False     True  False
3    False    False   True
4    False     True  False
5    True    False  False
6    False    False   True
7    True    False  False
8    False     True  False
9    True    False  False

Updated dataset after One-Hot Encoding:
   France  Germany  Spain   Age   Salary Purchased
0    True    False  False  44.0  72000.0        No
1    False    False   True  27.0  48000.0       Yes
2    False     True  False  30.0  54000.0        No
3    False    False   True  38.0  61000.0        No
4    False     True  False  40.0  63778.0       Yes
5    True    False  False  35.0  58000.0       Yes
6    False    False   True  38.0  52000.0        No
7    True    False  False  48.0  79000.0       Yes
8    False     True  False  50.0  83000.0        No
9    True    False  False  37.0  67000.0       Yes

Updated Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9 Data
columns (total 6 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
0   France     10 non-null     bool
1   Germany    10 non-null     bool
2   Spain      10 non-null     bool
3   Age        10 non-null     float64
4   Salary     10 non-null     float64 5   Purchased  10 non-null     object dtypes: bool(3), float64(2), object(1)
    memory usage: 402.0+ bytes

Final Preprocessed Dataset:
   France  Germany  Spain   Age   Salary  Purchased
0    True    False  False  44.0  72000.0          0
1    False    False   True  27.0  48000.0          1
2    False     True  False  30.0  54000.0          0
3    False    False   True  38.0  61000.0          0
4    False     True  False  40.0  63778.0          1
5    True    False  False  35.0  58000.0          1
6    False    False   True  38.0  52000.0          0
7    True    False  False  48.0  79000.0          1
8    False     True  False  50.0  83000.0          0
9    True    False  False  37.0  67000.0          1
```

## Output:

```
C:\Users\vsros\AppData\Local\Temp\ipykernel_20308\1860359933.py:40: FutureWarning: Downcasting behavior in `replace` is
updated_dataset['Purchased'] = updated_dataset.Purchased.replace(['No', 'Yes'], [0,
```